

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools : Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Scenario-Based Assessment

Topic 1: Footprinting Tools – Conducting Competitive Intelligence

Q1. (L3 – Apply) – 15 Marks

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named ABC Technologies. The consultant must collect publicly available information without directly interacting with the company's servers.

Task: Apply suitable footprinting tools to identify the company's:

- domain registration details,
- employee email addresses,
- technology stack,
- and publicly exposed subdomains.

Explain how each selected tool helps in conducting competitive intelligence.

Scenario

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named **ABC Technologies**. The consultant must gather publicly available information without directly interacting with the company's servers. The objective is to identify domain registration details, employee email addresses, technology stack, and publicly exposed subdomains.

Tools Used

- **WHOIS** – Retrieves domain registration information.
- **theHarvester** – Collects publicly available email addresses and host information.
- **WhatWeb** – Identifies the technologies used by a website.
- **Sublist3r** – Discovers publicly exposed subdomains.

How It Helps

- Reveals domain ownership and DNS information.
- Identifies publicly available employee email addresses.
- Detects web technologies and software used by the organization.

- Finds subdomains that may increase the organization's attack surface during security assessments.

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kali@kali: /usr/share/wordlists/metasploit
$ whois tcs.com
clear

Domain Name: TCS.COM
Registry Domain ID: 646354_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.networksolutions.com
Registrar URL: http://networksolutions.com
Updated Date: 2025-10-23T19:55:41Z
Creation Date: 1997-11-25T05:00:00Z
Registry Expiry Date: 2031-11-24T05:00:00Z
Registrar: Network Solutions, LLC
Registrar IANA ID: 2
Registrar Abuse Contact Email: domain.operations@web.com
Registrar Abuse Contact Phone: +1.877.228.6662
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: NS3.TCS.COM
Name Server: NS5.TCS.COM
Name Server: NS6.TCS.COM
Name Server: NS7.TCS.COM
Name Server: NS8.TCS.COM
DNSSEC: signedDelegation
DNSSEC DS Data: 61792 8 2 70BC14110DF8DA49862A63518BF982D3E41653F57A6E31AEC79C7D440F3D966
DNSSEC DS Data: 23191 8 2 76AD00F318BA7FCD1379F38C80685996670197E1DA415180BACE098FF7075460
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-28T08:14:43Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for

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Topic 2: DNS Zone Transfers

Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

Scenario

During an authorized penetration test, a security analyst suspects that the organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers (AXFR). The analyst performs a DNS zone transfer test to verify whether the server leaks DNS records.

Tools Used

- **dig** (DNS Information Groper)
- **dnsutils** package (contains the dig utility)

How It Helps

- Checks whether DNS zone transfers are permitted.
- Retrieves DNS records if the server is misconfigured.
- Identifies internal hosts, mail servers, and subdomains.
- Helps map the organization's network infrastructure for security analysis.
- Allows administrators to identify and correct DNS misconfigurations before attackers can exploit them.

A screenshot of a Kali Linux terminal window. The terminal shows the user at the prompt '(kali@kali) ~/usr/share/wordlists/metasploit'. They run the command 'dig -x', which returns 'DIG 9.20.15-2-Debian'. Then they run 'dig NS example.com', which returns a detailed DNS response for NS records of example.com, showing servers like 'elliott.ns.cloudflare.com' and 'hera.ns.cloudflare.com'. Finally, they run 'dig AXFR example.com @ns1.example.com', which returns an error: 'dig: couldn't get address for 'ns1.example.com': not found'.

Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

Q3. (L3 – Apply) – 20 Marks

Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

Scenario

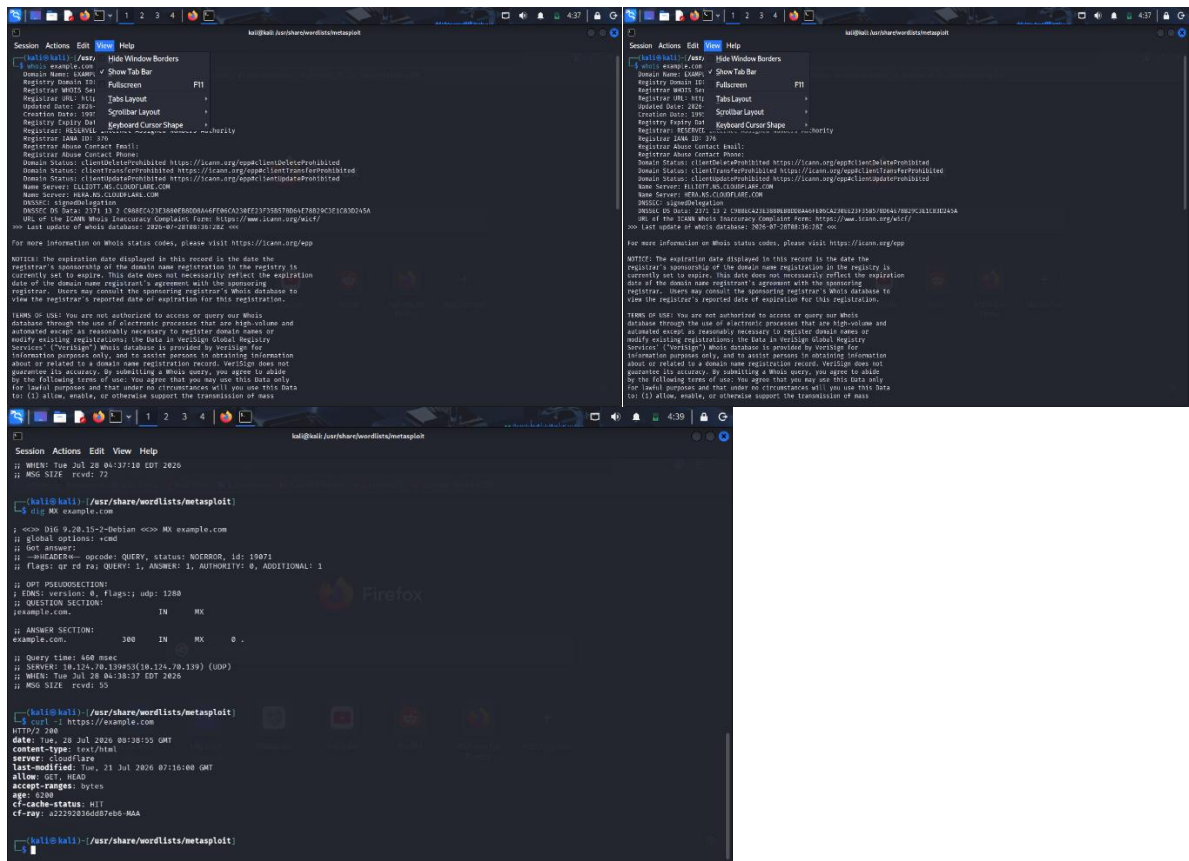
Employees receive an email claiming to be from the IT department, requesting them to verify their login credentials through a provided link due to an urgent security update. The objective is to identify the attack and apply suitable countermeasures before responding.

Tools Used

- **WHOIS** – Verifies the ownership of the domain in suspicious links.
- **dig / nslookup** – Retrieves DNS information for the suspicious domain.
- **curl** – Examines website HTTP response headers.
- **OpenSSL** – Checks the SSL/TLS certificate of the website.

How It Helps

- Verifies whether the website belongs to the legitimate organization.
- Detects suspicious or newly registered domains.
- Confirms whether the website uses a valid SSL certificate.
- Helps employees avoid phishing websites and credential theft.
- Encourages verification of suspicious emails before entering sensitive information or clicking links.



Annexure A

Scenario-Based

Code of Conduct:

I Anlo Melwin.S 192511299(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Anlo Melwin.S

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand